

Psychophysiological and behavioral engagement in Immersive Virtual Reality: Study of Visual and Auditory Stimuli Focused on Motor Rehabilitation Tasks

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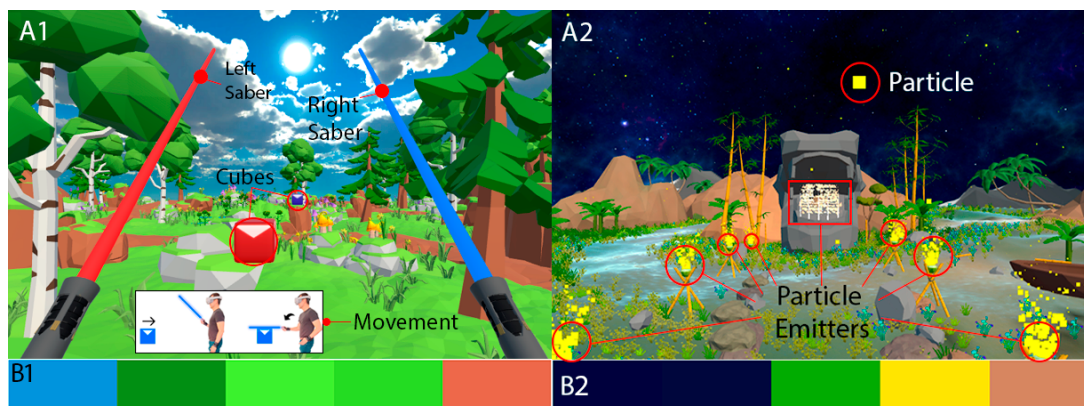


Figure 1: VR environment variations for low (A1) and high (A2) visual stimulation. B1 and B2 are the palette selections.

ABSTRACT

This study investigates the impact of multimodal stimuli in immersive virtual reality (VR) on emotional and psychomotor responses during an upper-limb task. An immersive VR game with two levels of visual and auditory stimuli allowed a 2x2 experiment. Forty-four participants completed the task while psychophysiological signals (EMG, ECG, EDA) and hand kinematics were recorded. Results showed that auditory stimulation significantly affected electrodermal responses and hand velocity, whereas visual stimulation reported no significant changes. These findings highlight the role of auditory cues in perceptual and autonomic modulation during VR interaction and their potential for designing immersive and rehabilitation-oriented VR systems.

Index Terms: Virtual reality, Virtual environments, Emotional responses, PhysiStimuli, Patient rehabilitation.

1 INTRODUCTION

Conventional therapies for upper-limb motor rehabilitation seek to improve mobility and reduce pain, and their effectiveness increases when emotional engagement is enhanced [5]. VR provides controlled environments that enhance functional performance and emotional engagement [6]. In this context, visual and auditory cues influence perception and attention, which in turn modulate motor behavior and promote performance and user engagement [7].

Visual and auditory stimuli help eliciting specific emotional states. For instance, Low-intensity visual stimuli promote baseline engagement, whereas high-intensity visual cues increase arousal and presence [4]. Similarly, low-intensity auditory stimuli support

sustained attention without overload, whereas high-intensity auditory cues enhance emotional activation and immersion. Combining both modalities allows the identification of configurations that maximize engagement, enabling changes in user perception and emotional and behavioral engagement [2].

This study uses task-oriented interactive VR scenarios with two visual and auditory levels to shape four affective-oriented conditions (HA, LA, HV, LV). Emotional engagement is assessed using electromyography (EMG), electrodermal activity (EDA), and electrocardiography (ECG) signals, which serve as objective indicators of emotional engagement. In contrast, behavioral engagement is evaluated through game performance metrics, including game score, hand velocity, and surface electromyography (sEMG).

2 METHODOLOGY

2.1 Design and Development of the Immersive Virtual Environments

The environments were built in Unity (v6) using Meta Quest 3s and OpenXR. Participants executed elbow flexion-extension movements to hit blocks synchronized to music (Fig. 1). Two visual conditions were designed: A) a low-stimulation condition using green hues associated with calm and low arousal, and B) a high-stimulation condition dominated by intense yellow tones linked to increased arousal. A blue background was included, as blue has been consistently associated with comfort and positive affect [11].

In condition A, visual complexity was intentionally minimized to promote a relaxation environment [8]: no particle systems were used, VR objects were limited to a slow motion and no animations were present. In contrast, condition B implemented high visual complexity by populating the scene with visual stimuli via the Unity's particle system (simultaneous instantiation of thousands of particles) [3]. Regarding audio, condition A used the track Rip and Tear from DOOM (2016) due to its high-intensity acoustic features [9], whereas condition B used Ludlow by the James Jackson

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Jazz Trio, selected for its low-tempo calming profile [10].

2.2 Experimental Design

Forty-four (44) participants (21 ± 3 years; 18 males, 26 females) performed four consecutive stages, each designed to comprehensively collect data related to participants' experiences:

(1) Introduction and informed consent, including the provision of instructions, signing of the informed consent form, and completion of a demographic questionnaire; (2) Instrumentation, where participants received instructions on the biosignals measurement device and were instrumented with EMG, ECG, EDA sensors, and wristbands; (3) Safety and experimentation, comprising safety instructions, level selection, baseline (resting-state) measurement, a tutorial, and execution of the experimental condition; and (4) Evaluation, involving verification of data acquisition, completion of the SAM questionnaire, and a final questionnaire.

2.3 Data Acquisition and Processing

ECG and EDA were recorded using a BiosignalsPlux device (Sampling rate = 1 kHz). ECG electrodes were placed on the sternum, V2, and V6; EDA electrodes on the non-dominant hand index and middle fingers. Signals were filtered, smoothed, and baseline-normalized to extract 19 features to compare between the experimental conditions, as suggested by Caldas et al. [1].

Performance was assessed through final score, hand velocity, and sEMG features. sEMG features were acquired from the dominant-hand biceps brachii muscle after filtering and normalizing the signal to the Maximum Voluntary Contraction (MVC). Signals were filtered and normalized to MVC, and fourteen EMG features were extracted. The root mean square (RMS) of hand velocity and descriptive metrics were extracted after resampling to 100 Hz and processing the signals from the Quest controllers' accelerometers.

3 STATISTICAL ANALYSIS

Analyses were performed in R (v4.4.1) on fifty-one (51) psychophysiological and behavioral features. A two-way ANOVA ($\alpha < 0.05$) was used to test the main effects of auditory and visual conditions when data were normally distributed, as assessed by the Shapiro-Wilk normality test. For non-normally distributed data, a non-parametric Kruskal-Wallis test was applied.

4 RESULTS

Data from two participants were excluded due to missing measurements. Ten participants per condition were analyzed. Significant differences were primarily observed in features related to the EDA signal and motor behavior, as reflected by hand velocity. In contrast, visual stimuli did not produce significant effects in any of the analyzed variables, and no significant interactions between auditory and visual stimuli were detected.

Figure 2 shows the standard deviation of the tonic EDA component and the mean Y-velocity of the dominant hand tend to be higher under high auditory stimulation, regardless of visual stimulation. However, unlike hand velocity, the EDA standard deviation decreases when visual stimulation is high. Low and high stimulus conditions were coded as 0 and 1, respectively.

Only one sEMG features (Max Peak Value - MPV) showed significance for auditory stimulation, but the lack of consistency across other sEMG metrics suggests it is not conclusive.

The EDA signal was analyzed under conditions of high and low auditory stimulation. Higher amplitude levels were observed during periods of increased auditory stimulation.

No significant effects were identified in features associated with the ECG signal or with the performance index.

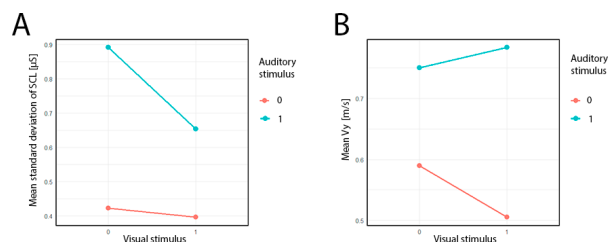


Figure 2: Standard deviation (A) and Mean average of the Y-axis component of hand velocity (B).

5 CONCLUSIONS

Auditory stimulation significantly influenced autonomic and motor responses by affecting EDA and hand velocity, whereas visual stimulation had no significant effects. Although performance remained unchanged, high-intensity audio elicited higher physiological activation and faster movements, while low-intensity audio promoted more controlled execution. The lack of effects in ECG and sEMG data may reflect a low-demanding task, due to the blocks' fixed speed. Overall, auditory cues play a stronger role than visual cues in simple VR motor tasks. Future work should incorporate more complex tasks, dynamic visual stimuli, and larger samples.

6 ACKNOWLEDMENT

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